

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

SE 19-206-0302 CHEMICAL PROCESS PRINCIPLES

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Exemplify material balance for various unit operations or processes with or without chemical reaction.
 CO2: Solve energy balance on processes with and without reaction.
 CO3: Explain the properties of thermodynamic systems and interpret the significance of the laws of thermodynamics.
 CO4: Summarize the working principle of the major equipments for screening, size reduction, and filtration.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	The potential energy of a body at a height of 15 m is 2 kJ. If the body is moving at a velocity of 50 m/s what is the kinetic energy ($g = 9.8067 \text{ m/s}^2$).	3	L1	1	1
	(b)	A wet stock of Ammonium sulphate containing 20% water is sent to a drier. The material leaving the drier contains 2.44% moisture. Determine how many kg of water is removed per kg of wet material charged. Also find the percent of original water in the feed that is removed by drying.	3	L2	1	3
	(c)	Derive the relation between enthalpy and internal energy.	3	L2	2	2
	(d)	State the law of conservation of energy.	3	L2	2	2
	(e)	Explain Hess's Law.	3	L1	3	2
	(f)	A gas mixture analyzing 20% (mol) CO, 30% (mol) H ₂ and 50% (mol) N ₂ is completely burns in air. The following heat of reaction data is available: $\text{CO(g)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} \quad \Delta H_r^\circ = -282.91 \text{ kJ/mol}$ $\text{H}_2\text{(g)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(g)} \quad \Delta H_r^\circ = -241.83 \text{ kJ/mol}$ Determine the amount of heat liberated on complete combustion of 100 mol of gas mixture.	3	L2	3	3
	(g)	Write notes on crushing rolls.	3	L2	4	1
	(h)	Explain about Belt conveyor.	3	L1	4	1

(P.T.O.)

BTS-III(R/S)-11-23-2147

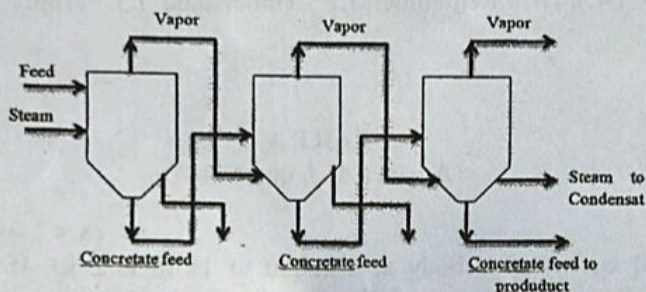
PART B

(4 × 12 = 48)

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| II. (a) Calculate enthalpy change for one mole of gas when it is heated from 400 K to 1500 K at 1 atm pressure given that the mean specific heat of the gas at the reference temperature of 273 K are 35 kJ/kmolK at 400 K and 50 kJ/kmolK at 1500 K. | 6 | L2 | 1 | 1 |
| (b) State: | 6 | L3 | 1 | 2 |
| (i) Daltons Law | | | | |
| (ii) Amagat's Law | | | | |
| (iii) Henry's Law. | | | | |

OR

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|---|---|----|---|---|
| III. (a) A triple effect evaporator is used to concentrate 100 kg of aqueous solution from a concentration of 20% solute to 80% solute. Assuming an equal amount of vaporization in each effect, calculate the composition and weight of the solution entering the second and third effect. | 8 | L3 | 1 | 3 |
|---|---|----|---|---|



- | | | | | |
|--|----|----|---|---|
| (b) Give a note on Mean Heat Capacity. | 4 | L2 | 1 | 2 |
| IV. The heat capacity of chlorine at 101.3 kPa varies with temperature as follows:
Calculate the enthalpy change when chlorine gas is heated from 273 K to 1273 K at 101.3 kPa. | 12 | L3 | 2 | 2 |

T (K)	273	373	473	573	673	773	873	973	1073	1173	1273
C _p (kJ/kmol K)	33.6	35.1	36.0	36.6	37.0	37.3	37.5	37.6	37.7	37.8	37.9

OR

- | | | | | |
|--|---|----|---|---|
| V. (a) A manufacturing plant released flue with this following composition. CO ₂ = 10.00%, O ₂ = 6.3%, CO = 2%, N ₂ = 81.70%. The heat capacity of the gases varies with temperature as given below where CP is in kJ/kmol K and T is in K. | 9 | L4 | 2 | 1 |
|--|---|----|---|---|

$$\text{O}_2: \text{CP} = 25.74 + 12.987 \times 10^{-3}T - 3.864 \times 10^{-6}T^2$$

$$\text{N}_2: \text{CP} = 27.03 + 5.815 \times 10^{-3}T - 0.289 \times 10^{-6}T^2$$

$$\text{CO}_2: \text{CP} = 26.540 + 42.454 \times 10^{-3}T - 14.298 \times 10^{-6}T^2$$

$$\text{CO}: \text{CP} = 26.586 + 7.582 \times 10^{-3}T - 1.12 \times 10^{-6}T^2$$

Calculate the enthalpy of the gas mixture at 750 K with reference to a base temperature of 298 K.

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|-------------------------|---|----|---|---|
| (b) Define Kopp's Rule. | 3 | L1 | 2 | 1 |
|-------------------------|---|----|---|---|

(Continued)

BTS-III(R/S)-11-23-2147

		Marks	BL	CO	PO
VI.	(a) Derive Energy Balance for a mechanical system.	8	L3	3	3
	(b) State Zeroth law of thermodynamics.	4	L2	3	1
OR					
VII.	(a) Describe about first law of thermodynamics for a Cyclic process.	6	L3	3	1
	(b) Calculate the heat of reaction for the esterification of ethyl alcohol with acetic acid if the standard heat of combustion is: Ethyl alcohol -1366.91 kJ/mol Acetic acid -871.69 kJ/mol Ethyl acetate -2274.48 kJ/mol	6	L2	3	2
VIII.	(a) Compare Jaw crusher and gyratory crusher with neat sketch.	8	L3	4	1
	(b) Discuss about flow through fluid, explain the significance of streamlining.	4	L3	4	2
OR					
IX.	(a) Define Filtration Theory.	3	L2	4	1
	(b) Elucidate on any three equipment of filtration.	6	L3	4	3
	(c) Write the significance filter aids.	3	L1	4	2

Bloom's Taxonomy Levels

L1 = 11.66%, L2 = 31.66%, L3 = 49.16%, L4 = 7.5%.
